

EXTENDING THE DATASET

Urban Risk-Aware Navigation via VQA-Based Event Maps

for People with Low Vision

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IRI (CSIC-UPC), Barcelona · [arXiv:2605.11782](https://arxiv.org/abs/2605.11782) · under revision at IEEE T-ITS

TODAY

Three parts

01

The paper

Why the problem matters and what we propose

02

How it works

From street images to risk maps, and how well it performs

03

Your task

Extending the dataset with new Mapillary sequences

Motivation

253M

people living with visual impairment in 2015: 36M blind, 217M with moderate to severe vision loss

703M

projected by 2050, driven by population ageing and growth

Obstacles, unmarked stairs, vehicles, crowds and construction make cities hard to walk alone.

One detector per hazard doesn't scale

Current approach: specialised pipelines

- A stair detector, a crosswalk analyser, ...
- Each needs its own data, training and tuning
- Misses hazards nobody built a model for

Proposed approach: VQA

- One Vision-Language Model answers questions in plain language
- A new hazard is a new question, with no retraining
- Answers become a map for route planning

What the paper contributes

Hierarchical questions

8 hazard categories, with follow-up questions in three levels

Risk score and event maps

Yes/No answers become a weighted risk for every street segment

A global dataset

20 cities on 6 continents, 820 labelled images. **This is what you will extend.**

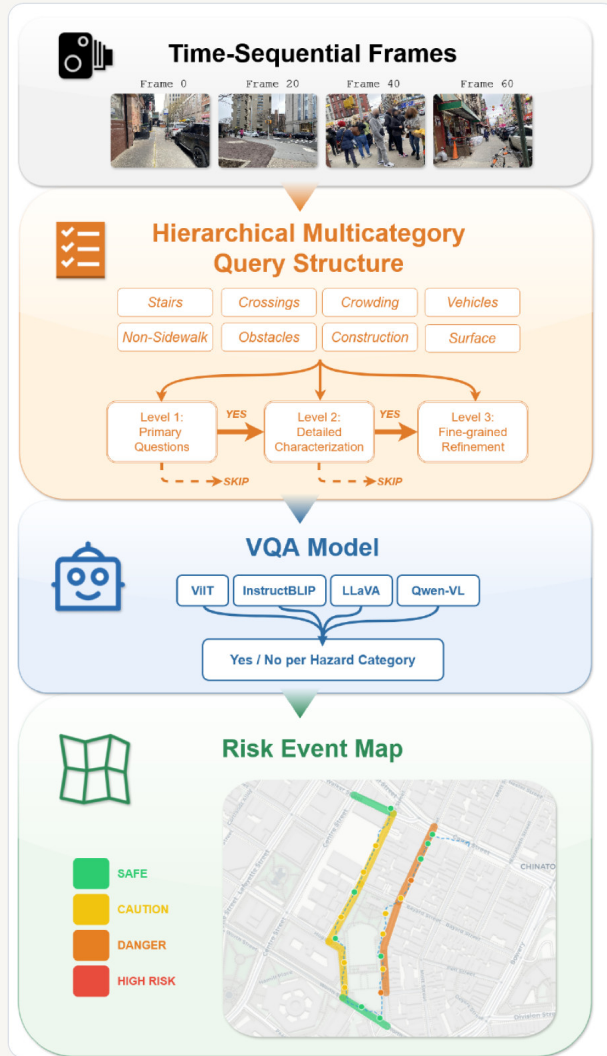
A model benchmark

ViLT, InstructBLIP, LLaVA and Qwen-VL compared on the same data

PART 2

How it works

From a walking video to a risk map



PART 2 · HOW IT WORKS

The pipeline

1 Keyframes

Images sampled along a walking route

2 Hierarchical questions

Asked level by level, only where relevant

3 VQA model

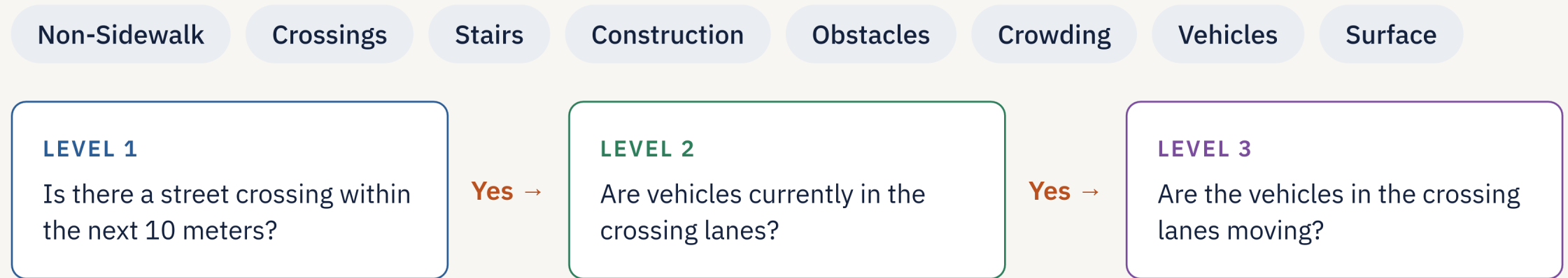
Answers Yes or No to each question

4 Risk event map

Answers become a risk score, drawn on the streets

PART 2 · HOW IT WORKS

Questions in three levels

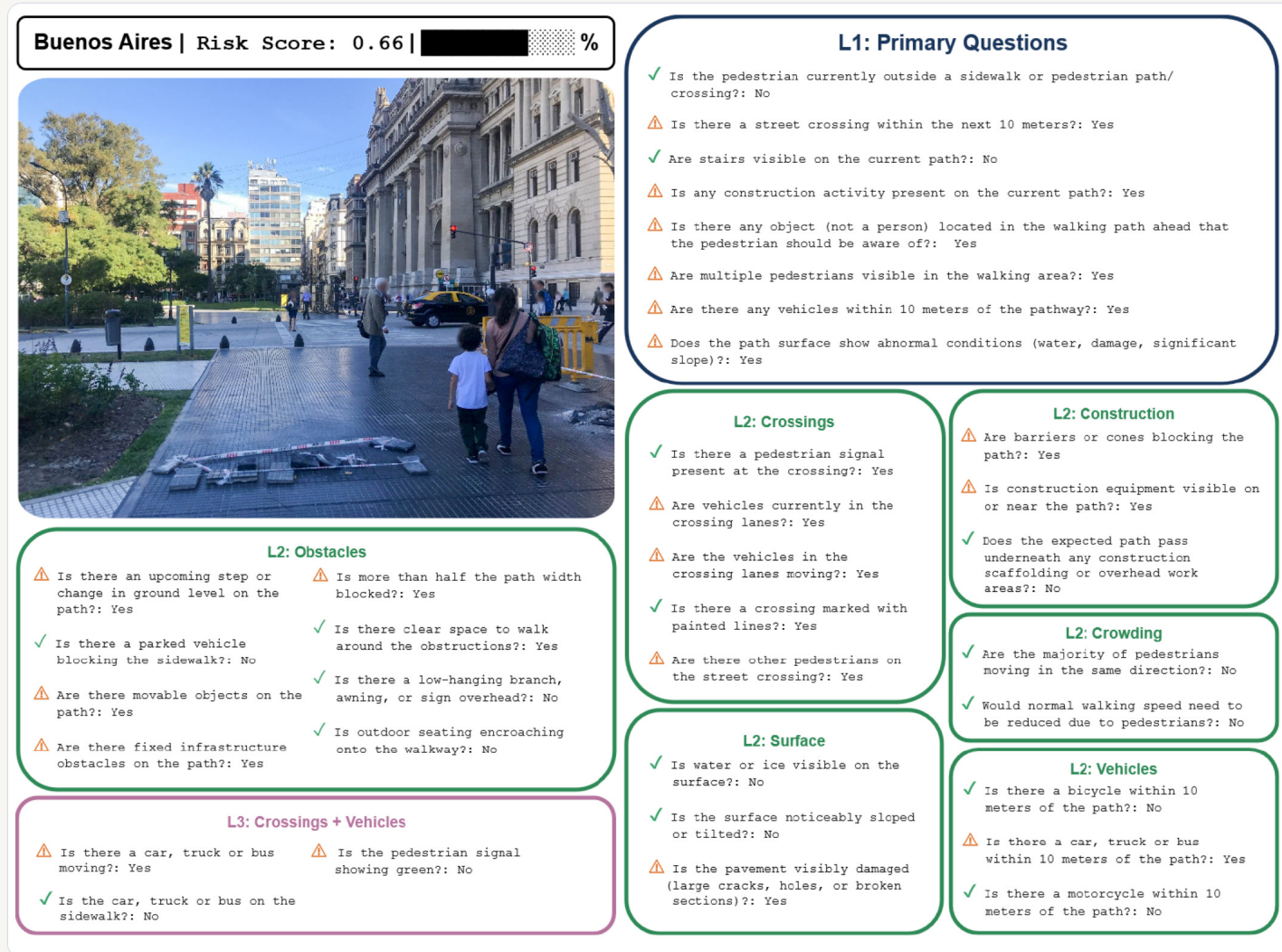


44 questions in total, but a **No** skips the whole branch. Every question starts with: *"You are an expert at detecting pedestrian obstacles for people with low vision. Answer only with Yes or No."*

PART 2 · HOW IT WORKS

One image, all levels

- Buenos Aires street
- Level 1 in blue, Level 2 in green, Level 3 in purple
- Warning signs: hazards. Check marks: safe
- Resulting risk score: **0.66**



PART 2 · HOW IT WORKS

Four models, two families

CLASSIFICATION

ViLT

Picks an answer from a fixed vocabulary and gives a confidence

GENERATIVE (MULTIMODAL LLMS)

InstructBLIP

Flan-T5-XL

LLaVA-1.6

Mistral-7B

Qwen2-VL

7B Instruct

Write the answer as text, conditioned on the image and the question

No model was retrained: all four are used as they come, with the same prompt.

All four run from `vqa/run.py`

PART 2 · HOW IT WORKS

From answers to a risk score

$$R = \max(0, \Sigma w \cdot x) / \Sigma w$$

- Only the 8 Level-1 questions count
- $x = 1$ when the answer is Yes (hazard)
- $x = -1/8$ when it is No (a small safety reward)
- Result between 0 (safe) and 1 (every hazard present)

Tier	w	Categories
Critical	1.0	Construction, Surface, Non-Sidewalk
High	0.6	Crossings, Stairs, Obstacles
Low	0.3	Crowding, Vehicles



PART 2 · HOW IT WORKS

Risk event maps

- Each keyframe snaps to its nearest OpenStreetMap street
- A street takes the **max** risk of its images: it is only as safe as its most dangerous point

Safe ≤ 0.15

Caution < 0.4

Danger < 0.7

High risk ≥ 0.7

Grey streets: no images, not "safe".

PART 2 · HOW IT WORKS

The dataset

20

cities on 6 continents

820

images labelled by hand

Sources: Mapillary (crowd-sourced street imagery) and our own recordings in Barcelona with Biel Glasses smart glasses. 20 keyframes per sequence.

41

sequences

18,352

answered questions



Paper, Fig. 3

PART 2 · RESULTS

Qwen-VL gives the best balance

Model	Accuracy	F1	Precision	Recall	Specificity	MAE_R
InstructBLIP	0.70	0.45	0.69	0.34	0.91	0.12
LLaVA	0.73	0.55	0.68	0.49	0.86	0.14
Qwen-VL (best)	0.77	0.69	0.68	0.70	0.81	0.14
ViLT	0.33	0.47	0.33	0.87	0.04	0.76

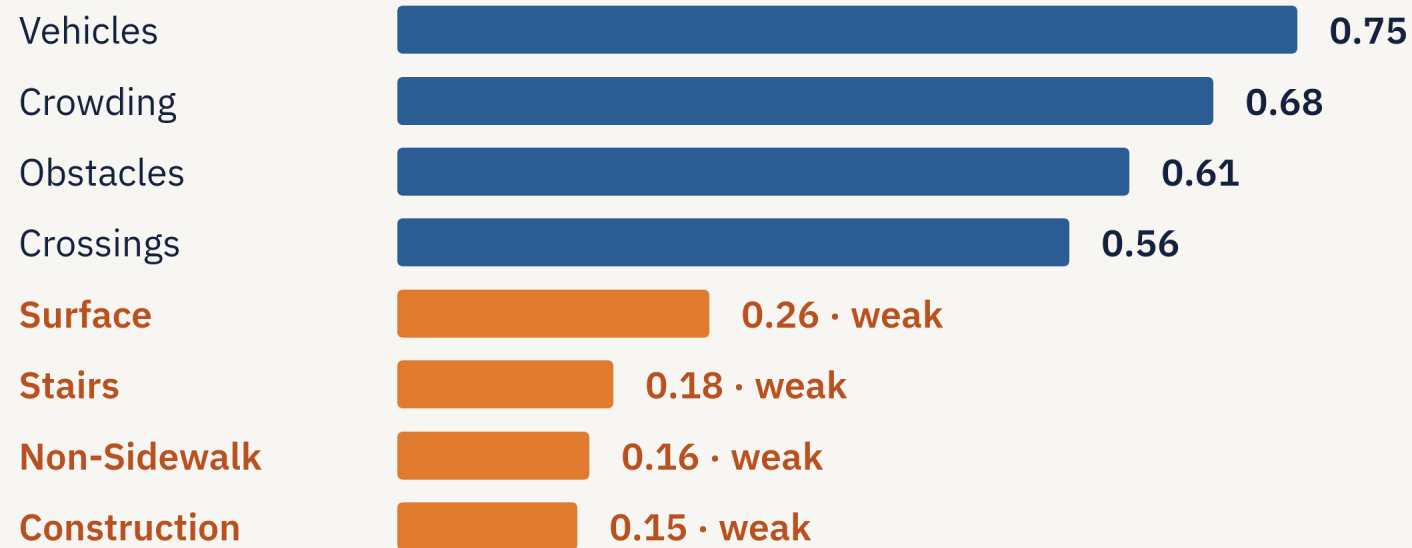
ViLT flags almost everything as a hazard (specificity 0.04)

InstructBLIP misses two thirds of the hazards (recall 0.34)

Qwen-VL balances both: F1 0.69, accuracy 0.77

PART 2 · RESULTS

Safety-critical categories have the lowest F1



The weak categories have **few positive examples** in the dataset. More data for them is the most useful thing we can add.

PART 3

Your task

Extending the dataset with new sequences

PART 3 · YOUR TASK

Add new Mapillary sequences

Look especially for

Stairs

Construction works

Damaged or wet surfaces

People walking outside the sidewalk

Cities and countries not covered yet

PART 3 · YOUR TASK

Workflow

1

Choose

`mapillary.com/app`

2

Download

`download_mapillary.py`

Images and GPS positions

3

Label

`label_gt.py`

Pick 20 keyframes and
answer the questions

4

Hand in

`pack_data.sh`

Send me the data

We then run the models and review the labels. The exact commands are in **README §5, "Adding a new Mapillary sequence"**.

All scripts are in the `dataset/` folder of the repository

PART 3 · STEP 1

What makes a good sequence

- ✓ **Pedestrian point of view:** walking, camera facing forward.
No car or dashcam
- ✓ **Good GPS, as far as you can tell:** the track on the map follows the street, with no jumps
- ✓ **Adds diversity:** a new city, country or kind of street
- ✓ **Not 360°:** panoramas are rejected automatically
- ✓ **Enough clear frames:** at least ~100 images, sharp and in daylight
- ✓ **Shows weak categories:** stairs, works, damaged surfaces

In the Mapillary viewer, the **capture key** is the sequence ID. An image ID from the URL (pKey=...) works too.

PART 3 · STEP 3

The labelling tool

Key	Action
Y / N	Yes / No
B	Back to the previous Level-1 question
S	Skip this image (asked again next run)
R	Repeat the previous image's answers
Q	Quit (the image in progress is lost)

A · Pick 20 keyframes

The route is split into 20 parts. In each, press **K** to keep a sharp, forward-facing image or **S** to see the next one.

B · Answer the questions

One image and one question at a time. Follow-ups appear after a Yes. Finished images are saved; run it again to continue.

Labelling rules

Pedestrian point of view

Answer from the camera's point of view, looking forward

"Nearby" means about 10 m

For crossings and vehicles

Only what is visible

Not what you know or guess about the place

Be consistent

Setup

Setup (README §2)

1. Clone the repository
2. Python 3.10 virtual environment, then install requirements.txt
3. Download the dataset archive (link in the README)
4. Copy .env.example to .env and add your own free Mapillary token

RESOURCES

Everything is in the README

github.com/AntoniValls/VQA-for-event-maps

§2 Setup

Environment, tokens, dataset

§5 Adding a sequence

Every step, command by command

§11 Gotchas

Common mistakes to avoid

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